

GenMol / UDLM study through V13

34 configurations; 68 scheduled seed runs; 5,504 planned requests; 5,504 independently rescored accepted requests. Status: complete. No superiority or formal promotion is established.

V13 tests temperature placement using the same frozen CE checkpoint within each empirical/MASK-rich prior pair. Both modes use T=0.5, 128 predictor network evaluations per molecule, top-p=1, no Gibbs, and seeds 2100/2101 with 100 requests each. Raw mode converts CE to LOO before tempering; clean mode tempers CE logits first, converts to LOO, and uses bridge temperature one.

Four V13 settings

Configuration / accepted	Scoring	Quality % $\pm$ SD	Diversity $\pm$ SD
e_ce_raw_t050 200/200; {'completed': 2}	repaired	46.500 $\pm$ 2.121	0.893 $\pm$ 0.005
e_ce_raw_t050 200/200; {'completed': 2}	strict	34.500 $\pm$ 2.121	0.878 $\pm$ 0.002
e_ce_clean_t050 200/200; {'completed': 2}	repaired	52.000 $\pm$ 9.899	0.891 $\pm$ 0.001
e_ce_clean_t050 200/200; {'completed': 2}	strict	34.500 $\pm$ 6.364	0.876 $\pm$ 0.002
mask_ce_raw_t050 200/200; {'completed': 2}	repaired	49.500 $\pm$ 3.536	0.882 $\pm$ 0.001
mask_ce_raw_t050 200/200; {'completed': 2}	strict	42.000 $\pm$ 2.828	0.873 $\pm$ 0.003
mask_ce_clean_t050 200/200; {'completed': 2}	repaired	47.500 $\pm$ 2.121	0.892 $\pm$ 0.009
mask_ce_clean_t050 200/200; {'completed': 2}	strict	32.500 $\pm$ 0.707	0.878 $\pm$ 0.005

Both declared contrasts: clean temperature minus raw-LOO temperature

Entries below are signed within-seed differences followed by their equal-seed mean and sample SD. Quality is in percentage points; diversity is in raw units. Primary is empirical CE; secondary is MASK-rich CE. Pairing is by seed, not matched molecular trajectories.

Contrast / scoring	Metric	Seed 2100; 2101	Mean $\pm$ sample SD
primary / repaired	quality	+0.000; +11.000	+5.500 $\pm$ 7.778
primary / repaired	diversity	-0.007; +0.003	-0.002 $\pm$ 0.006
primary / strict	quality	-3.000; +3.000	+0.000 $\pm$ 4.243
primary / strict	diversity	-0.005; +0.002	-0.002 $\pm$ 0.005
secondary / repaired	quality	-6.000; +2.000	-2.000 $\pm$ 5.657
secondary / repaired	diversity	+0.017; +0.003	+0.010 $\pm$ 0.010
secondary / strict	quality	-12.000; -7.000	-9.500 $\pm$ 3.536
secondary / strict	diversity	+0.007; +0.004	+0.006 $\pm$ 0.001

Definitions and limits

Quality = within-seed unique valid molecules with QED ≥ 0.6 and SA ≤ 4 / all requested samples. QED measures drug-likeness (higher favored); SA estimates synthetic accessibility (lower favored). Validity = valid/requested; uniqueness = unique valid/valid. Diversity uses the released PyTDC metric on unique valid molecules. Repaired scoring applies SAFE repair and keeps the largest component; strict SAFE disables repair. Both remove tokenizer special tokens before decoding, so exact final control-token counts are recorded separately in overview.json.

Every $\pm$ value is sample SD across two seeds, not a confidence interval. Missing or failed seeds withhold configuration and paired means/SD; retained partial source aggregates are explicitly separate. All four validity/uniqueness/quality/diversity outcomes and pair differences remain in the untouched V13 appendix and JSON/CSV.

These are adaptive engineering pilots across 34 settings, not independent confirmatory tests. V5/V6 use 128 requests per setting; V9/V10/V12/V13 use 200. The local MDLM reference uses $3 \times 1,000$ requests, repaired quality 85.8%; strict quality is not directly comparable to that repaired reference. The baseline's 3,000 requests and failed V4 diagnostic's 32 requests are outside UDLM accepted totals. Different study seeds cannot support paired cross-study effects. Final UDLM seeds 0/1/2 remain reserved.

V13 adds no training. Historical R/S/E adaptation used $1,000 \text{ updates} \times \text{batch16} = 16,000$ exposures. Empirical CE and MASK-rich CE each used a fresh MDLM 50k EMA initialization, $1,000 \text{ updates} \times \text{batch128} = 128,000$ exposures, seed1500; the changed prior changes denoising difficulty. V8 CT's failed post-exit controller and separate checkpoint audit, unstarted original CE arm, separate successful V8b CE, V11 prelaunch failure with no trained checkpoint, and successful V11b MASK CE remain preserved. Inference costs are not uniform across all studies: V10 includes 512 NFE, four times 128. V13's planned molecule-NFE budget is 102,400.

Appendix order: untouched terminal V13 report, then the preserved 78-page overview through V12 (including its original reports, training/configuration details, prior/checkpoint identities, failures, and historical caveats). Older appendices retain the statements current when they were written. All original bytes, source relocations, inputs, and output hashes are recorded in the new manifest.